

ARTIN (ALIREZA) KAVOOSI M.Sc.

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Blog: Artin Kavoosi

Address: Iran, Tehran

EDUCATION

University of Tehran

Tehran, Iran

M.Sc. in System Modeling and Data Analytics

Sep 2023 - Present

- GPA: 18.74/20, (4/4)
- Supervisor: Prof. Tavakkoli-Moghaddam
- Thesis: "Towards autonomous supply chains for perishable products: Learning-based decision-making for coordination, competition, and resilience"
- (expected completion: Sep 2026)

Kharazmi University

Tehran, Iran

B.S. in Industrial Engineering

Sep 2019 - Jun 2023

- GPA: 16.44/20

RESEARCH INTERESTS

Methods: Operations research, Machine learning, Convex optimization

Domain: Logistics, Inventory management, Supply chain design

PUBLICATIONS

Published papers

1. Kavoosi, A., Tavakkoli-Moghaddam, R., Sajedi, H., Tajik, N., & Tafakkori, K. (2025) Dynamic pricing and inventory control of perishable products by a deep reinforcement learning algorithm. *Expert Systems with Applications*, 291, 128570.

Working and under review papers

1. Kavoosi, A., Tavakkoli-Moghaddam, R., Ziari, M., & Azaron, A. Coordinating pricing and replenishment in a blood supply chain with deep reinforcement learning. Under review. *Transportation Research Part E*.
2. Kavoosi, A., Tavakkoli-Moghaddam, R. Disruption-resilient blood supply chain by a structural causal graph dynamics and decentralized reinforcement learning approach. Under review. *Computers & Industrial Engineering*.
3. Kavoosi, A., Tavakkoli-Moghaddam, R. Cross-Entropy Neural Networks for Mixed Strategies in a Finite-Horizon Supplier-Retailer Inventory Duopoly with Lead Times and Rationing. Submitted.
4. Kavoosi, A., Tavakkoli-Moghaddam, R. Online Learning of Neural Ordinary Differential Equations for Dynamic Inventory Control Under Non-Stationary Demand. Submitted.

ACADEMIC EXPERIENCE

Research assistant | University of Tehran

Sep 2024 - Present

- Studying and researching on Inventory control and supply chain problems, supervised by Prof. Tavakkoli-Moghaddam.

Teaching assistant | University of Tehran

Mar 2024 - Sep 2025

- Machine learning (Spring 2025, Instructor: Prof. Jamili)
- Macroeconomics (Spring 2025, Instructor: Prof. Ghaderi)
- Soft computing (Fall 2024, Instructor: Prof. Tavakkoli-Moghaddam)
- Linear algebra (Spring 2024, Instructor: Prof. Rahbari)

Teaching Assistant | Kharazmi University

Mar 2023 - Jan 2023

- Engineering mathematics II (Spring 2023, Instructor: Prof. Khamse)

WORK EXPERIENCE	Business Partner Family Fashion Boutiques (Privately Owned) Jun 2020 - Present • Co-manage three family-owned fashion boutiques (Aylar & Tirajeh) serving customers across Tehran Province, Iran. • Oversee daily store operations, including inventory management, merchandising, and sales activities. • Deliver high-quality customer service while supporting business growth and maintaining long-term customer relationships.	
	Research Intern Sapco Company Apr 2022 - Oct 2022 • On data-driven transportation cost estimation, application of SAP tools and software to optimize inventory operations, and analyzing workflows using BPMN. • SAPCO is an Iranian automotive parts supply and procurement company that manages the sourcing and quality of components for Iran Khodro, one of Iran's largest automotive manufacturers.	
SERVICE	Reviewer for <i>Transportation Research Part E: Logistics and Transportation Review</i>	
AWARDS AND HONORS	• Top 0.4% among +10,000, University Master's Degree Entrance Exam (Konkour) in Iran, Industrial Engineering 2023 • Top 1% among +150,000, Nationwide University Entrance Exam (Konkour) in Iran, Mathematics and Physics 2019	
	Areas: Data analytics, Markov decision processes, Linear and integer programming. Software: Python, Julia, GAMS, IBM SPSS, LaTeX. Languages: Persian (Native), Turkish (Native), English (Professional).	
REFERENCES	Prof. Reza Tavakkoli-Moghaddam School of Industrial Engineering University of Tehran tavakoli@ut.ac.ir	Dr. Amir Azaron Sauder School of Business University of British Columbia amir.azaron@ubc.ca
	Dr. Nazanin Tajik Department of Industrial and Systems Engineering Mississippi State University tajik@ise.msstate.edu	Dr. Matineh Ziari School of Industrial Engineering University of Tehran m.ziari@ut.ac.ir